

8-Bit Successive-Approximation Register ADC

on GF180MCU PDK
Team Hungry Chippos

PDK	GF180MCU (GF180MCUD open-source)
Track	Track A – Analog / Mixed-Signal
Target Resolution	8 bits
Target Speed	≥ 1 MS/s (stretch: 10 MS/s)
Supply Voltage	3.3 V (I/O domain)
Reference Voltage	External V_REF (1.65 V)
Clock	External (no on-chip oscillator)
GitHub Repo	github.com/brubru6707/hungry-chippos-chipathon-2026

Executive Summary

The Hungry Chippos propose to design, simulate, lay out, and tape out a fully open-source 8-bit Successive-Approximation Register (SAR) Analog-to-Digital Converter (ADC) targeting the GF180MCU process design kit. The design targets a minimum sampling rate of 1 MS/s (with a stretch goal of 10 MS/s) at a 3.3 V supply with an external reference voltage, and is architected as a charge-redistribution SAR using a binary-weighted capacitor DAC and a regenerative dynamic comparator.

The primary contribution is a reproducible, fully open-source analog design flow — from hierarchical Xschem schematics through ngspice corner simulation to Magic / KLayout layout, LVS, and DRC sign-off — with every step scripted and documented so that a third party can clone the repository and reproduce the simulation results with a single command inside the IIC-OSIC-TOOLS Docker container.

The design is deliberately scoped to be achievable by a six-person team within the Chipathon timeline, with a pre-agreed descope ladder that gracefully degrades from 8-bit / 10 MS/s to 6-bit / 1 MS/s if schedule or layout constraints force it. Key explicit out-of-scope items — on-chip clock generation, on-chip bandgap reference, and full post-silicon characterisation — are called out upfront so the contribution is tractable and the reviewers can evaluate realism.

1. Background and Motivation

1.1 Why a SAR ADC?

SAR ADCs occupy a well-understood Pareto frontier in the speed–power–area design space: at 8-bit resolution and megasample-per-second throughput they require no continuous-time residue amplifier, minimise static power, and map cleanly onto the capacitor and switch primitives available in a mature digital-biased process like GF180MCU. For a six-person team with a four-month design window, a SAR ADC offers a clear functional specification, published design equations, measurable test vectors (DNL, INL, ENOB), and a natural decomposition into independently workable sub-blocks.

Importantly, a SAR ADC is an unambiguous ground truth for whether the design is correct: a correct 8-bit converter passes all 256 code transitions within ± 0.5 LSB DNL and ± 1 LSB INL. This makes it straightforward to define binary pass/fail verification gates at each design stage, and to detect whether a layout change has degraded the functional result — a discipline directly analogous to the verification-gated philosophy praised in Track D peer review.

1.2 The Two Problems We Are Not Confusing

This proposal designs and delivers a functional ADC. It explicitly does not claim to address on-chip voltage reference design (a separate bandgap problem), on-chip clock synthesis, or post-silicon calibration of systematic offset. Each of those would be a project in itself; stating this upfront keeps the contribution tractable and lets reviewers assess realism rather than scope.

2. Project Scope

Scope statement: Design, simulate, lay out, and tape out a charge-redistribution 8-bit SAR ADC in GF180MCU, verified against pre-registered DNL / INL / ENOB acceptance criteria across process corners, with a fully reproducible open-source simulation and layout flow.

2.1 In Scope (Deliverables)

- **Primary analog blocks:**
 - Dynamic latch comparator — low-offset regenerative topology (Emily, Luc)
 - Binary-weighted charge-redistribution capacitor DAC (**Max, Mimi**)
 - SAR digital controller — synchronous FSM in Verilog, synthesised and placed with OpenROAD (**Sam**)
 - Top-level integration, bootstrap switch / input sampling network, padding (Bruno)
- **Verification suite:**
 - ngspice transient simulations across TT, FF, SS process corners
 - Monte Carlo offset and mismatch analysis for the comparator
 - DNL and INL extraction from the simulated transfer curve (code-density or ramp method)
 - ENOB and SNDR from FFT of a simulated sine-wave input
- **Layout (Magic + KLayout):**
 - Common-centroid capacitor array with dummy fringe caps
 - Matched comparator layout; guard rings on all sensitive nodes
 - DRC-clean and LVS-clean sign-off against GF180MCU rules
 - Integration into chipathon-supplied padding template

- **Reproducibility package:** single-command Docker simulation reproduce (see §4).
- **Workshop paper:** architecture, design choices, simulated and measured results.

2.2 Out of Scope (Deferred / Future Work)

- On-chip clock generation (PLL or ring oscillator) — external clock assumed.
- On-chip bandgap voltage reference — external V_{REF} pin assumed.
- Post-silicon calibration of residual offset or gain error.
- Analog test infrastructure (scan chain, BIST for ADC) — out-of-scope for v1.
- Differential input mode — single-ended input in v1.
- Technology porting to SKY130 or other PDKs — GF180MCU only.

Each item above is a genuine extension that could be a follow-up project. Naming them here prevents scope creep and signals to reviewers that the team has a realistic model of the design space.

2.3 Scope Freeze Rule

The architecture, verification targets, and descope ladder lock at the end of Phase 1. Post-freeze ideas go into BACKLOG.md in the repository. Any mid-flight scope change requires a written justification committed to the repository.

3. Technical Approach

3.1 Architecture Overview

The converter uses charge-redistribution architecture, the dominant topology for SAR ADCs at 8-bit resolution. On every conversion cycle the input voltage is sampled onto the top plates of the capacitor array; the bottom plates are then switched by the SAR logic to perform a binary search. After $N = 8$ iterations the digital output word $B[7:0]$ is valid. The five functional blocks and their interfaces are:

- Input sampling network — bootstrap switch for improved linearity at full-rate input; feeds top plate of cap array.
- Binary-weighted capacitor DAC — total capacitance $256 C_u$; C_u sized for kT/C noise floor below quantisation noise floor at 8 bits.
- Dynamic latch comparator — StrongARM or double-tail topology; self-timed; output drives SAR logic.
- SAR digital controller — synchronous FSM clocked by external clock; generates DAC switching sequence; outputs $B[7:0]$ and EOC.
- Pading and I/O — chipathon shared pading template; VDD, GND, V_{IN}, V_{REF}, CLK, $D[7:0]$, EOC pads.

3.2 Comparator Design (Emily, Luc)

A regenerative dynamic latch comparator is chosen for its zero static power and self-timed operation. The target specification is an input-referred offset of less than half an LSB (approximately 6.5 mV at $V_{REF} = 1.65$ V) at the 3σ level across the TT corner. Monte Carlo analysis will be used to verify this; if the TT result shows $\sigma_{offset} > 2$ mV, the device sizes will be increased to recover margin before tapeout.

Luc's primary role is corner analysis: the comparator must meet the offset target across FF, SS, and TT corners, and across the supply tolerance band (3.0 V to 3.6 V). Failure to meet the SS corner target triggers the 8→6-bit descope (item 2 in the descope ladder).

3.3 Capacitor DAC (Sam, Mimi)

The binary-weighted array uses MIM capacitors available in GF180MCU. The unit capacitor C_u is sized to place the thermal noise floor (kT/C noise referred to the input) at least 6 dB below the 8-bit quantisation noise floor, giving $C_u \geq 50$ fF. Total array capacitance is $256 C_u \approx 13$ pF.

Mimi leads layout, implementing a common-centroid arrangement with identical dummy cells at the array periphery to minimise systematic gradient mismatch. The acceptance criterion is $DNL < 0.5$ LSB at the schematic level (pre-layout); post-layout target relaxes to $DNL < 1$ LSB with DRC/LVS clean.

3.4 SAR Digital Controller (Max)

A standard synchronous FSM is implemented in synthesisable Verilog, targeting a clocked design that does not depend on the comparator's regeneration time for correct operation (the comparator output is registered on a defined clock edge). The controller is synthesised using Yosys and placed-and-routed with OpenROAD into a small digital macro, which is then integrated into the top level. Timing closure is targeted at 20 MHz (sufficient for 1 MS/s with $N = 8$ plus overhead cycles), with the stretch goal at 200 MHz for 10 MS/s.

3.5 Verification Strategy

All simulation pass/fail thresholds are pre-registered in VERIFICATION_PLAN.md (Phase 1 deliverable). A block that does not pass its registered gate is not allowed to proceed to layout. This is the analog equivalent of a software verification oracle.

The gates are:

- Gate 1 — Comparator schematic TT: $\sigma_{\text{offset}} < 2$ mV (Monte Carlo $N \geq 200$ runs); propagation delay < 2 ns at $V_{\text{IN}} = V_{\text{CM}} \pm 5$ mV.
- Gate 2 — DAC schematic TT: settling to 0.5 LSB accuracy within 40 ns for all 256 codes.
- Gate 3 — Top-level schematic TT: $DNL < 0.5$ LSB, $INL < 1$ LSB, $ENOB \geq 7.5$ bits (from sine FFT).
- Gate 4 — Corner sweep (FF, SS): all metrics degrade by less than 1 bit ENOB; $DNL < 1$ LSB.
- Gate 5 — Post-layout LVS + DRC: zero errors; post-layout simulation $ENOB \geq 7.0$ bits at TT.

Gates 1–3 must pass before layout begins on that sub-block. Gate 4 triggers the descope decision if the SS corner fails. Gate 5 is the tapeout sign-off condition.

4. Deliverables

Deliverable	Description
Schematic + simulation suite	Full hierarchical schematic in Xschem; ngspice testbenches for all blocks and top-level; swept across process corners
Layout (Magic + KLayout)	DRC/LVS-clean layout of comparator, cap array, SAR logic, padding integration in GF180MCU
Tapeout package (GDSII)	Sign-off GDSII integrated into chipathon shared padding; DRC/LVS logs attached
Verification plan	VERIFICATION_PLAN.md: pre-registered pass/fail gates (DNL, INL, ENOB) per block and per corner

Deliverable	Description
Reproducibility package	Docker image SHA, PDK commit, tool versions, ngspice netlists; single-command simulation reproduce
Workshop paper	Architecture, design choices, measurement results (simulated + silicon where available)

4.1 Reproducibility Manifest

A REPRODUCIBILITY.md file committed with every tapeout-candidate commit will specify:

- IIC-OSIC-TOOLS Docker image SHA256
- GF180MCU PDK commit hash
- Xschem, ngspice, Magic, KLayout, Yosys, OpenROAD version strings
- Shell command to reproduce all simulation results from a clean clone

The intent is that a reviewer can run a single shell script and obtain the same DNL/INL plot and ENOB number reported in the workshop paper. Closed-source EDA tools are not used; the entire flow runs inside the open IIC-OSIC-TOOLS container.

5. Timeline

Phase	Period	Key Deliverables
Phase 1	May 2026	Block architecture locked; schematic stubs committed; testbench harness scaffolded; toolchain verified; descope ladder documented; verification plan pre-registered in VERIFICATION_PLAN.md
Phase 2	May–June 2026	All blocks at schematic-complete + ngspice TT-corner pass; comparator offset Monte Carlo; cap array unit-cell layout; SAR RTL simulating; reproducibility scripts drafted
Phase 3	July 2026	Full schematic simulation across FF/SS/TT corners (DNL, INL, ENOB, SNDR); layout started (comparator + cap array); LVS clean on sub-blocks; mentor design review with placeholder GDSII
Phase 4	Aug–Sep 2026	Layout complete; DRC/LVS clean; padding integration; tapeout package submitted; workshop paper draft
Phase 5	Post-tapeout	Silicon bring-up; measured ENOB / SNDR reported; open repository updated

6. Pre-Agreed Descope Ladder

If schedule slips or corner analysis reveals a fundamental block-level limit, the team drops items in the following order. Each descope is triggered by a specific, observable event, not by a subjective judgment call.

#	Descoped Item	Condition that triggers descope
1	10 MS/s speed target → relax to 1 MS/s	Comparator or DAC settling prevents meeting 10 MS/s across corners by end of Phase 3
2	8-bit resolution → 6-bit resolution	Post-layout DNL > 1 LSB and cannot be fixed by routing iteration by Phase 4 start
3	On-chip sample-and-hold buffer	If bootstrap switch proves infeasible in GF180MCU, use external held input
4	SNDR / SFDR characterisation	Silicon bring-up risk; simulation result reported if silicon unavailable
5	Unhardened baseline comparator (control)	Drop if schedule pressure forces it; primary ENOB result stands alone

The final fallback that is never removed: at minimum, a DRC/LVS-clean GDSII of a 6-bit, 1 MS/s converter with simulated results is submitted. The hardened pipeline, simulation harness, and reproducibility package remain in the repository regardless of the resolution or speed achieved.

7. Risks and Mitigations

Risk	Likelihood	Mitigation
Comparator offset > ½ LSB at TT corner	Medium	Sized for $\leq 3\sigma$ offset < ½ LSB; if corner analysis fails, relax speed to 1 MS/s (descope)
Cap array mismatch degrades ENOB to < 7 bits	Medium	Common-centroid layout + dummy caps; if DNL > 1 LSB post-layout, drop to 6-bit target
DAC settling time limits speed target	Medium-High	Descscope to 1 MS/s if 10 MS/s target cannot meet spec across all corners
SAR logic timing closure slips	Low	Standard-cell P&R with generous timing margin; decouple from analog schedule
Tapeout area exceeds chipathon share (< ~2235 × 2235 μm^2)	Low	Estimated core area ~0.1 mm^2 ; well within quota; monitor iteratively
LVS failures in padding integration	Low-Medium	Use chipathon-supplied padding template; integrate early in Phase 3
Team member unavailability	Low-Medium	Paired ownership on every block (Emily/Luc, Max/Mimi); no single point of failure

8. Team

Member	Role	Responsibility
Bruno	Team Lead / Integration	Top-level integration, padding, I/O, final GDSII assembly
Bruno	Comparator	Dynamic comparator schematic, simulation, layout
Luc	Comparator (Co-lead)	Comparator offset calibration, corner analysis
Sam	SAR Logic	Binary-weighted cap array design, matching analysis
Mimi	DAC / Cap Array (Co-lead)	DAC layout, common-centroid arrangement, DRC/LVS
Max	DAC / Cap Array	Digital SAR controller in Verilog, synthesis, P&R

Each primary sub-block has two owners, eliminating single points of failure. Bruno as integration lead is responsible for ensuring sub-block interfaces (schematic pins, Pydantic-equivalent pin contracts documented in the repository README) are agreed at Phase 1 and do not change without a written update.

9. Repository Structure

The repository layout below reflects the reproducibility-first philosophy. All simulation scripts are in scripts/; all results land in results/ and are never committed by hand — they are generated by the scripts.

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hungry-chippos-chipathon-2026/
  README.md — Quick-start: one-command reproduce
  REPRODUCIBILITY.md — Docker SHA, PDK commit, tool versions
  VERIFICATION_PLAN.md — Pre-registered pass/fail gates (Phase 1)
  BACKLOG.md — Post-freeze ideas
  comparator/
    schematic/ — Xschem .sch files
    sim/ — ngspice testbenches + corner scripts
    layout/ — Magic .mag files
  dac/
    schematic/
    sim/
    layout/
  sar_logic/
    rtl/ — Verilog FSM
    syn/ — Yosys synthesis scripts
    pnr/ — OpenROAD P&R scripts
  top/
    schematic/ — Top-level Xschem integration
    sim/ — Top-level DNL/INL/ENOB testbenches
    layout/ — Top-level Magic + padding
  scripts/
    run_all_sims.sh — Entry point: reproduces all results
    extract_dnl_inl.py — Post-process ngspice output → DNL/INL/ENOB
    monte_carlo.sh — Automated MC sweep for comparator offset
  results/ — Auto-generated; gitignored (except CI artefacts)
  tapeout/
    gds/ — Final GDSII
    drc_logs/
    lvs_logs/

```

10. References

The following works inform the design and methodology of this proposal.

- <https://github.com/sscs-ose/sscs-chipathon-2026/blob/main/resources/Analog/eda/README.md>
- https://labs.engineering.asu.edu/mixedsignals/wp-content/uploads/sites/58/2023/04/Low-Power_SAR_ADC_Design_Overview_and_Survey_of_State-of-the-Art_Techniques-1.pdf
- <https://github.com/SamS581/hungry-chippos-chipathon-2026>
- https://en.wikipedia.org/wiki/Successive-approximation_ADC
- <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10704295>
- Optimize speed, resolution, and power

<https://techschems.com/sar-adc-schematic>

https://pages.jh.edu/aandreo1/216/Archives/2014/Handouts/POP_Ch3-4.pdf

- Logic gates from NFETs and PFETs + Rules

Example Designs

<https://www.yilectronics.com/Students/tsnakai/Labs/Project/Project.html>

- Make NAND → make D flip flop → 8-bit SAR
- Tested at each step

https://www.ee.columbia.edu/~kinget/EE6350_S16/03_ADC_Dhruv_Gurkaran_Shikhar/schematic.html

- 16 flip flops (8 shift, 8 control)

<https://gemini.google.com/share/b2b6cdf181c7>

<https://share.google/aimode/s1aaorTjUL1QYcPYM>

- Option 1: xschem with multiple shift register+ring counters
- Option 2: Verilog with coded block

https://www.yilectronics.com/Courses/ENGR338_CE/f2021/proj/SARADC_LTSpiceReport.pdf

Design

<https://notebooklm.google.com/notebook/a845c4f2-c32e-43c8-9e55-b57e1f3be735?authuser=2>

- <https://www.falstad.com/circuit/>
- https://www.seas.ucla.edu/brweb/papers/Journals/BR_Magazine4.pdf
- Razavi, B. Principles of Data Conversion System Design. IEEE Press, 1995. — Foundational SAR ADC architecture and noise analysis.
- Ginsburg, B.P., Chandrakasan, A.P. '500-MS/s 5-bit ADC in 65-nm CMOS.' IEEE JSSC, 2006. — Charge-redistribution SAR reference.
- van Elzakker, M., et al. 'A 10-bit charge-redistribution ADC consuming 1.9 μ W at 1 MS/s.' IEEE JSSC, 2010. — Low-power SAR design reference.
- IIC-OSIC-TOOLS: Open-Source IC Design Environment. <https://github.com/iic-jku/IIC-OSIC-TOOLS> — Chipathon-mandated tool container.
- GF180MCU PDK. <https://github.com/google/gf180mcu-pdk> — Process design kit.
- OpenROAD Project. <https://theopenroadproject.org> — Digital P&R flow for SAR logic.
- IEEE SSCS PICO Chipathon 2026 Guidelines — Contest rules, area limits, padding template.
- Bootstrap Switch: (emily will format later)
- <https://ieeexplore.ieee.org/document/9531736> - 5) Lin-Hsieh Bootstrapped Switch
- https://www.researchgate.net/publication/2983201_Switch_Bootstrapping_for_Precise_Sampling_Beyond_Supply_Voltage
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Contact and Links

GitHub Repository	github.com/brubru6707/hungry-chippos-chipathon-2026
Progress Tracker	https://github.com/brubru6707/hungry-chippos-chipathon-2026/blob/main/PROGRESS.md
GDS File	TBD — to be committed by Phase 4 tapeout deadline
Pin List	TBD — to be registered in Phase 1 (VDD, GND, V_IN, V_REF, CLK, D[7:0], EOC)
Team Contact	brubru6707

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